Amlan Banaji – Academic CV

MSCA Postdoctoral Fellow in Mathematics, University of Jyväskylä, Finland
Email: banajimath@gmail.com
Research area: Fractal geometry, harmonic analysis and dimension theory
Website | ORCID | Google Scholar

Research Interests

My research lies at the intersection of fractal geometry, harmonic analysis, and dimension theory. I study Fourier decay and refined notions of dimension for measures and attractors generated by conformal and non-conformal iterated function systems. My recent work establishes polynomial Fourier decay for broad classes of fractal measures and develops dimension spectra that capture scaling behaviour beyond classical Hausdorff and box dimensions. These questions also connect with number theory, probability, and dynamical systems.

Employment History

University of Jyväskylä
September 2025 – present
Marie Skłodowska-Curie Actions Postdoctoral Fellow in Mathematics
End date: 31 August 2027. Working on the project "Fourier decay for nonlinear fractal measures" (FoDeNoF)

University of Jyväskylä
March 2025 – August 2025
Postdoctoral Researcher in Mathematics
Working with Tuomas Orponen on fractal geometry and dimension theory

Loughborough University
March 2023 – February 2025
Research Associate in Mathematics
Working with Simon Baker as a postdoc on fractal geometry and Fourier analysis

Publications and Preprints

Preprints:

16. A. Banaji, H. Yu. *Fourier transform of nonlinear images of self-similar measures: qualitative aspects.* arXiv:2606.09743

15. A. Banaji, H. Chen, A. Rutar, W. Wang. Attainable forms of lower spectra. arXiv:2603.03174

14. S. Baker, A. Banaji. *Self-similar and self-conformal measures with slow Fourier decay. arXiv:2602.05593*

13. S. Baker, A. Banaji, D.-J. Feng, C.-K. Lai, Y. Xiong. *Distinct dimensions for attractors of bi-Lipschitz iterated function systems.* arXiv:2509.22084

12. A. Banaji, A. Rutar. *Lower box dimension of infinitely generated self-conformal sets.* arXiv:2406.12821

Published and accepted articles:

11. A. Banaji, H. Yu. *Fourier transform of nonlinear images of self-similar measures: quantitative aspects,* **Peking Mathematical Journal** (to appear). arXiv:2503.07508

10. A. Banaji, J. M. Fraser, I. Kolossváry, A. Rutar. *Assouad spectrum of Gatzouras–Lalley carpets,* **Advances in Mathematics** 484 (2026), Article 110707. arXiv:2401.07168, DOI

9. S. Baker, A. Banaji. *Polynomial Fourier decay for fractal measures and their pushforwards,* **Mathematische Annalen** 392 (2025), 209–261. arXiv:2401.01241, DOI

8. A. Banaji, A. Rutar, S. Troscheit. *Interpolating with generalized Assouad dimensions,* **Journal of Geometric Analysis** 35 (2025), Article 270. arXiv:2308.12975, DOI

7. A. Banaji, I. Kolossváry. *Intermediate dimensions of Bedford–McMullen carpets with applications to Lipschitz equivalence,*
Advances in Mathematics 449 (2024), Article 109735. arXiv:2111.05625, DOI

6. A. Banaji, J. M. Fraser. *Assouad type dimensions of infinitely generated self-conformal sets,*
Nonlinearity 37 (2024), Article 045004. arXiv:2207.11611, DOI

5. A. Banaji. *Generalised intermediate dimensions,*
Monatshefte für Mathematik 202 (2023), 465–506. arXiv:2011.08613, DOI

4. A. Banaji. *Metric spaces where geodesics are never unique,*
American Mathematical Monthly 130 (2023), 747–754. arXiv:2209.00598, DOI

3. A. Banaji, J. M. Fraser. *Intermediate dimensions of infinitely generated attractors,*
Transactions of the American Mathematical Society 376 (2023), 2449–2479. arXiv:2104.15133, DOI

2. A. Banaji, H. Chen. *Dimensions of popcorn-like pyramid sets,*
Journal of Fractal Geometry 10 (2023), 151–169. arXiv:2212.06961, DOI

1. A. Banaji, A. Rutar. *Attainable forms of intermediate dimensions,*
Annales Fennici Mathematici 47 (2022), 939–960. arXiv:2111.14678, DOI

Grants and Awards

- *2025*: Awarded **€226,000 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship** with 100% score "Fourier decay for nonlinear fractal measures" (FoDeNoF), grant no. 101210409, University of Jyväskylä.
- *2023–2024:* Awarded **LMS Travel Grant** for Early Career Researchers to visit the University of Oulu
- *2019*: **Postgraduate Gray Prize** for the best MSc student in the Faculty of Science and Medicine at the University of St Andrews.

Teaching, Supervision and Research Leadership

- *June 2026* – present: **Bachelor's thesis supervision:** Ossi Paananen, University of Jyväskylä, primary supervisor.
- *March–May 2026*: Sole **Lecturer** for Probability Theory 2 (MATS262) at the University of Jyväskylä; responsible for lectures, exercise sessions, course materials, assessment, and marking; 28 hours of lectures and 14 hours of exercise sessions.
- *January–February 2026:* **Secured and hosted** a one-month University of Jyväskylä Visiting Fellowship for Haipeng Chen.
- *March–May 2024*: **lecturer** for second-year course 'Elements of Topology' to 90 students at Loughborough University, contributing to assessment and marking.
- *2019–2022*: Led undergraduate **tutorials** at the University of St Andrews: MT2502 Analysis (10 groups total), MT2505 Abstract Algebra (2 groups), MT1003 Pure and Applied Mathematics (2 groups).

Academic Service and Professional Memberships

- *2022–present*: **Referee** for more than 15 journals, including *IMRN, Advances in Mathematics, Mathematische Annalen, Journal of the LMS, Ergodic Theory and Dynamical Systems, and Nonlinearity;* reviewer for *MathSciNet*.
- *2026:* **Organiser** of a session on Fractal Geometry at the Finnish Mathematical Days 2026.
- *2023–2025:* **Co-organiser** of the Loughborough University Dynamical Systems Seminar
- *2024:* **Co-organiser** of the Workshop on Ergodic Theory and Fractal Geometry at Loughborough University
- *2022:* **Organiser** of St Andrews Analysis Reading Group
- *2021*: **Co-organiser** of the Postgraduate Interdisciplinary Mathematics Symposium (PIMS), St Andrews.
- **Professional memberships**: London Mathematical Society (LMS), Institute of Mathematics and its Applications, Edinburgh Mathematical Society.

Talks and Minicourses

Selected invited talks and minicourses (more than 50 talks in total; full list):

- **Minicourse**, Fourier decay of fractal measures
  Dimension theory of iterated function systems and Fourier decay of nonlinear self-similar measures workshop, Shenzhen Technology University, China, 25–26 June 2025
- **Seminar talk,** Fourier decay of nonlinear images of self-similar measures
  University of Helsinki Geometric and Functional Analysis Seminar, Finland, 17 April 2025
- **Seminar talk,** Coincidence and disparity of fractal dimensions for dynamically defined sets
  IMPAN Dynamical Systems seminar, Warsaw, Poland, 17 February 2025
- **Minicourse**, Fourier decay for nonlinear fractal measures
  Focused workshop on harmonic analysis methods in fractal geometry, Erdős Center, Budapest, Hungary, 4–8 November 2024
- **Keynote talk**, Overlapping iterated function systems
  British Early Career Mathematicians' Colloquium, Birmingham, UK, 14 June 2024
- **Seminar talk**, Fourier decay of fractal measures and their pushforwards
  University of Warwick Ergodic Theory and Dynamical Systems Seminar, UK, 7 November 2023
- **Invited online talk,** Dimensions of self-affine carpets. Slides. Video.
  One World Fractals (talks by myself and Hong Wang), 18 January 2023
- **Inaugural colloquium talk**, Intermediate dimensions. Slides
  Shenzhen Technology University Mathematics Colloquium, China (online), 22 October 2022

Outreach and Knowledge Exchange

- **Authored expository articles** in *American Mathematical Monthly* and *Newsletter of the LMS*
- Delivered **general audience talks** such as Loughborough Pint of Science 2024.
- Ran an **interactive mathematics stall** at the European Researchers' Night 2025.

Education

University of St Andrews
United Kingdom
2019–2023

PhD Mathematics
Thesis: "Interpolating between Hausdorff and box dimension"
Graduated 28 November 2023
Topic: Fractal geometry and dimension theory, with the Analysis Research Group
Supervisors: Jonathan Fraser (primary), Kenneth Falconer

University of St Andrews
United Kingdom
2018–2019

MSc Mathematics, Distinction
Overall MSc mark: 19.5/20
Dissertation: Solvability of Partial Differential Equations on Fractal Domains
(Score: 19.1/20, supervised by Professor Kenneth Falconer)

University of Cambridge
King's College
United Kingdom, 2015–2018

BA (Hons) Mathematics
Selected Part II courses: Linear Analysis, Analysis of Functions, Topics in Analysis, Differential Geometry, Riemann Surfaces, Logic and Set Theory